

Topic	Surface Hydrology	
Module	2	Curve number method

This module shows you how to use the NRCS (SCS) Curve number method to estimate runoff volume and peak runoff.

The NRCS (SCS) curve number method was originally devised by the Soil Conservation Service (SCS), which is now called the Natural Resources Conservation Service (NRCS) residing within the United States Department of Agriculture. The method was developed to compute water losses from rainfall events. Water losses are also called “abstractions” and they include infiltration, evapotranspiration, interception, depression storage, and other losses. Once losses are computed, the resulting water becomes runoff. The Curve Number (CN) method is described in this module for the calculation of runoff volume and peak runoff, but if an input rainfall hyetograph is used, the method allows for the computation of a resulting runoff flow hydrograph. Further details on how this is done can be obtained in various hydrology textbooks found in the library and in the NEH-4 (National Engineering Handbook – 4) publication (SCS 1985).

Estimating runoff volume

Estimations of total runoff volume from design storm events are needed for the construction of wetlands, retention ponds, flood control dams, and other water infrastructure. The total volume is also needed for estimating potential total maximum daily loads (TMDL) of pollutants being routed to surface waters and for developing guidelines for water quality in streams and rivers. The curve number (CN) method is ideally suited for estimating runoff volumes.

The curve number method assumes there are two components of water losses: an initial abstraction I_a and a continuing abstraction F_a (Figure 1). The runoff (or excess precipitation P_e) is then a result of the difference between the total precipitation and those components of water losses as presented in the following equation:

$$P_e = P - (I_a + F_a) \quad (1)$$

Where,

P = precipitation (mm)

P_e = excess precipitation or direct runoff (mm)

I_a = initial abstraction before ponding of water occurs (mm)

F_a = continuing abstraction, or the additional depth of water retained in a catchment. $F_a \leq S$, where S is some potential maximum water storage in the catchment.

Note: If P is less than I_a (initial abstraction) all your water will infiltrate, and P_e will be 0.

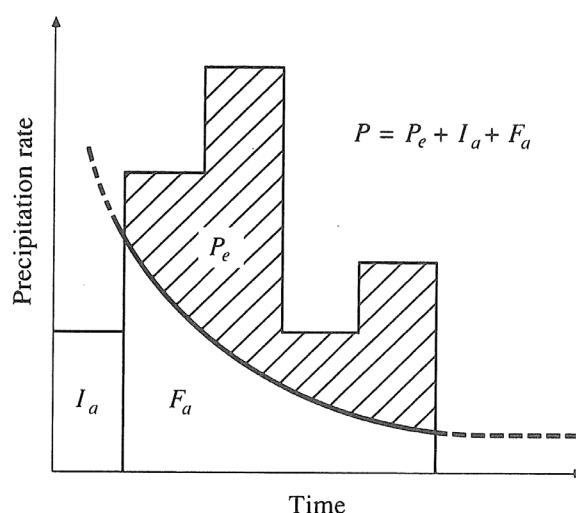


Figure 1. Total precipitation (P) in a storm event equal to excess precipitation (Pe) plus water losses (Ia and Fa). Source: Chow et al., 1988 and SCS, 1986.

The hypothesis behind the curve number method is that the ratios of the catchment storage values and the precipitation values are equal:

$$\frac{F_a}{S} = \frac{P_e}{P - I_a} \quad (2)$$

When we combine equations 1 and 2 we get that:

$$P_e = \frac{(P - I_a)^2}{(P - I_a) + S} \quad (3)$$

where

P_e	=	excess precipitation or runoff (mm),
P	=	rainfall (mm),
S	=	potential maximum retention after runoff begins (mm), and
I_a	=	initial abstraction (mm).

Initial abstraction (I_a) represents all losses before runoff begins. It includes water retained in surface depressions, water intercepted by vegetation, evaporation, and infiltration. I_a can be highly variable but generally is correlated with soil and cover parameters. Through studies of many small agricultural catchments, I_a was found to be approximated by the following empirical equation:

$$I_a = 0.2S \quad (4)$$

By removing I_a as an independent parameter, this approximation allows use of a combination of S and P to produce a unique runoff amount. Substituting equation 4 into equation 3 gives:

$$P_e = \frac{(P - 0.2S)^2}{(P + 0.8S)} \quad (5)$$

S is related to the soil and cover conditions of the catchment through the curve number (CN). CN has a range of 0 to 100, and S is related to CN by:

$$S = \frac{25400}{CN} - 254 \quad (6)$$

Notice that a high CN results in high imperviousness and thus greater potential for runoff. Low CN numbers are designated to highly pervious surfaces, resulting in less runoff.

The major factors that determine CN are the hydrologic soil group, cover type, treatment, hydrologic condition, and antecedent runoff condition. Another factor often considered is whether impervious areas outlet directly to the drainage system or whether the flow spreads over pervious areas before entering the drainage system. Tables 3-5 at the end of this module provide CN values for a range of soil and cover conditions.

Notice that the CN values for each land cover condition is defined by the 4 hydrological soil types as defined in Table 1.

Table 1. Definition of hydrological soil types.

Hydrological soil type	Description
A	Deep sand, deep loess, aggregated silts
B	Shallow loess, sandy loam
C	Clay loams, shallow sandy loam, soils low in organic matter, high in clay
D	Soils that swell when wet, heavy plastic clays and certain saline soils.

The CN factor can further be corrected for wet or dry antecedent moisture conditions (AMC). The curve number for dry conditions (AMC I) can be adjusted from the curve numbers presented in tables for normal conditions (AMC II) as follow:

$$CN(I) = \frac{4.2 \cdot CN(II)}{10 - 0.058 \cdot CN(II)} \quad (7)$$

For wet condition (AMC III), the following adjustment is made:

$$CN(III) = \frac{23 \cdot CN(II)}{10 + 0.13 \cdot CN(II)} \quad (8)$$

For catchments with multiple landuses or soil type areas, you would calculate the CN by a weighted averaging approach.

Example 1 – Runoff volume calculations for example catchment

Compute the runoff volume from a 20 mm storm event in a 100 ha catchment in the Banks Peninsula. The storm event occurred after very dry conditions. The land use in the catchment is:

- 30% residential of 0.1 ha lots
- 10% residential of 0.2 ha lots
- 10% paved roads with curbs and stormwater sewers
- 40% pasture in good condition
- 10% open spaces (parks, etc. in fair condition)

Ten years later the catchment becomes urbanized. Compute the volume of runoff for the same storm event given the following:

- 40% residential of 0.1 ha lots
- 30% residential of 0.2 ha lots
- 15% roads
- 5% gravel roads
- 10% parks (good condition)

Estimating peak runoff rate

The peak discharge in the curve number method is derived from a triangular approximation hydrograph of our excess precipitation of duration D.

This method was described by SCS (1989) and was originally developed for uniform rainfall. The time to peak flow is:

$$T_p = D/2 + T_L = D/2 + 0.6 T_c \quad (9)$$

where

$$\begin{aligned} T_p &= \text{time to peak (T),} \\ D &= \text{duration of excess rainfall (T),} \\ T_L &= \text{time of lag (T),} \\ T_c &= \text{time of concentration (T).} \end{aligned}$$

Time of concentration equal to $T_L/0.6$ is the longest travel time and may be obtained from the equation:

$$T_c = L^{0.8} \left[\frac{\left(\frac{1000}{CN} - 9 \right)^{0.7}}{4407 \cdot (S_g)^{0.5}} \right] \quad (10)$$

where

$$\begin{aligned} T_c &= \text{time of concentration (h),} \\ L &= \text{longest flow length in catchment (m),} \\ CN &= \text{runoff curve number,} \\ S_g &= \text{average catchment gradient (m/m).} \end{aligned}$$

The time of peak is necessary to develop a design hydrograph for routing runoff through a storage reservoir or for combining hydrographs from several subcatchments. It is not required for peak flow estimates. The peak flow rate is calculated from the following equation:

$$q = q_u A P_e F_p \quad (11)$$

where

$$\begin{aligned} q &= \text{peak runoff rate (m}^3\text{/s),} \\ q_u &= \text{unit peak flow rate (m}^3\text{/s per ha per mm of runoff),} \\ A &= \text{catchment area (ha),} \\ P_e &= \text{runoff depth (mm)} \\ F_p &= \text{pond and swamp adjustment factor (see Table 2)} \end{aligned}$$

The unit flow rate is obtained from graphs using the time of concentration and the ratio of initial abstraction to 24-hour rainfall (see Figure 2). This ratio represents the fraction of rainfall that occurs before runoff begins. The initial abstraction is usually taken as $0.2S$, where S is the maximum difference between rainfall and runoff calculated, as previously presented.

Table 2. Adjustment factor (F_p) for pond and swamp areas throughout the catchment.

Percentage of Pond and Swamp Areas	F_p
0.0	1.00
0.2	0.97
1.0	0.87
3.0	0.75
5.0	0.72
Source: SCS (1986).	

The curve number method is the widely used in the United States and other countries for estimating peak runoff rates on small drainage basins. In New Zealand, the curve number method has been adopted by the Auckland Regional Council (ARC, 1999) and is routinely used for stormwater management and design. Application of the curve number method is simple and direct; however, differences in results are caused by the choice of different procedures, especially for estimating T_c and by the choice of curve numbers.

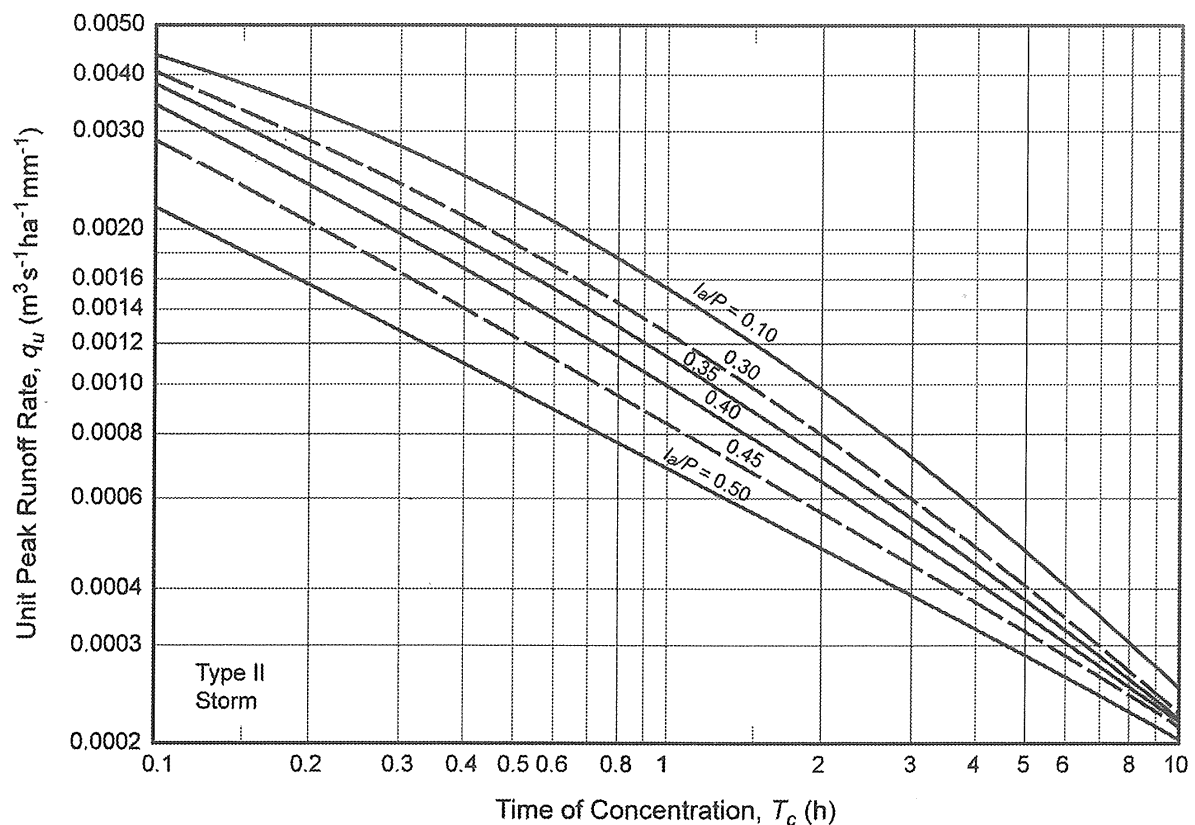


Figure 2. Unit peak runoff rates for a SCS type II rainfall distribution (Huffman et al., 2011 and SCS, 1990).

Example 2 – Peak runoff calculations for example catchment

Determine the peak runoff rate from a 100 ha catchment from a 120 mm, 24 h storm that produces 10 mm depth of runoff. Assume a flow length of 1500 m, AMC II, a weighted average curve number of 75, and average catchment gradient of 0.02 m/m, and 0.2% pond and swamp areas.

For basins up to 30 square kilometers, curve number procedures developed from TR-55 (SCS, 1986) may be used to predict peak runoff rates. Those procedures are described in the references.

References

ARC 1999. Guidelines for stormwater runoff modelling in the Auckland Region. Technical Publication 108 (TP 108). Auckland Regional Council.

Huffman, R.L., Fangmeier, D.D., Elliot, W.J., Workman, S.R., and Schwab, G.O. (2011). Soil and Water Conservation Engineering. Sixth edition. American Society of Agricultural and Biological Engineers.

Maidment, D.R. (ed). (1992). Handbook of Hydrology. McGraw-Hill, Inc., NY.

SCS (1986). Urban Hydrology for Small Catchments. TR-55, Soil Conservation Service, USDA, Washington, DC.

SCS (1990). Engineering Field Manual, Chapter 2. Soil Conservation Service. Washington, D.C.

Additionally, several computerized sources of the TR-55 method are available.

Table 3. Runoff curve number for urban areas with AMC(II) and Ia=0.2S.

Cover Description		Curve Numbers for Hydrologic Soil Group			
Cover Type and Hydrologic Condition	Average % Impervious Area ^[a]	A	B	C	D
Fully developed urban areas (vegetation established)					
Open space (lawns, parks, golf courses, cemeteries, etc.): ^[b]					
Poor condition (grass cover < 50%)		68	79	86	89
Fair condition (grass cover 50% to 75%)		49	69	79	84
Good condition (grass cover > 75%)		39	61	74	80
Impervious areas:					
Paved parking lots, roofs, driveways, etc. (excluding right-of-way)		98	98	98	98
Streets and roads:					
Paved; curbs and storm sewers (excluding right-of-way)		98	98	98	98
Paved; open ditches (including right-of-way)		83	89	92	93
Gravel (including right-of-way)		76	85	89	91
Dirt (including right-of-way)		72	82	87	89
Western desert urban areas:					
Natural desert landscaping (pervious areas only) ^[c]		63	77	85	88
Artificial desert landscaping (impervious weed barrier, desert shrub with 1- to 2-inch sand or gravel mulch and basin borders)		96	96	96	96
Urban districts:					
Commercial and business	85	89	92	94	95
Industrial	72	81	88	91	93
Residential districts by average lot size:					
0.05 ha or less (town houses)	65	77	85	90	92
0.1 ha	38	61	75	83	87
0.13ha	30	57	72	81	86
0.2 ha	25	54	70	80	85
0.4 ha	20	51	68	79	84
0.8 ha	12	46	65	77	82
Developing urban areas					
Newly graded areas (pervious areas only, no vegetation) ^[d]		77	86	91	94
^[a] The average percent impervious area shown was used to develop the composite CNs. Other assumptions are as follows: impervious areas are directly connected to the drainage system, impervious areas have a CN of 98, and pervious areas are considered equivalent to open space in good hydrologic condition. ^[b] CNs shown are equivalent to those of pasture. Composite CNs may be computed for other combinations of open space cover type. ^[c] Composite CNs for natural desert landscaping should be computed using procedures from SCS (1986) based on the impervious area percentage (CN = 98) and the pervious area CNs. The pervious area CNs are assumed equivalent to desert shrub in poor hydrologic condition. ^[d] Composite CN to use for the design of temporary measures during grading and construction should be computed using procedures from SCS (1986) based on the degree of development (impervious area percentage) and the CNs for the newly graded pervious areas. Adapted from SCS (1986).					

Table 4. Runoff Curve Number (CN) for agricultural lands with AMC(II) and $I_a=0.25$.

Land Use or Crop	Treatment or Practice ^[a]	Hydrologic Condition ^[b]	Curve Number for Hydrologic Soil Group			
			A	B	C	D
Fallow	Bare soil		77	86	91	94
	Crop residue (CR)	Poor	76	85	90	93
		Good	74	83	88	90
Row crops	Straight row (SR)	Poor	72	81	88	91
		Good	67	78	85	89
	SR + CR	Poor	71	80	87	90
		Good	64	75	82	85
	Contoured (C)	Poor	70	79	84	88
		Good	65	75	82	86
	C + CR	Poor	69	75	82	86
		Good	64	74	81	85
	C + Terraced (T)	Poor	66	74	80	82
		Good	62	71	78	81
	C + T + CR	Poor	65	73	79	81
		Good	61	70	77	80
Small grain	Straight row	Poor	65	76	84	88
		Good	63	75	83	87
	SR + CR	Poor	64	75	83	86
		Good	60	72	80	84
	Contoured	Poor	63	74	82	85
		Good	61	73	81	84
	C + T	Poor	61	72	79	82
		Good	59	70	78	81
	C + T + CR	Poor	60	71	78	81
		Good	58	69	77	80
Close-seeded legumes or rotation meadow	Straight row	Poor	66	77	85	89
		Good	58	72	81	85
	Contoured	Poor	64	75	83	85
		Good	55	69	78	83
	C + T	Poor	63	73	80	83
		Good	51	67	76	80

^[a] Crop residue cover applies only if residue is on at least 5% of the surface throughout the year.

^[b] Hydrologic condition is based on combination factors that affect infiltration and runoff, including (a) density and canopy of vegetative areas, (b) amount of year-round cover, (c) amount of grass or close-seeded legumes, (d) percent of residue cover on the land surface (good ≥ 20 %), and (e) degree of surface roughness.

Source: SCS (1972, 1986).

Table 5. Runoff curve number (CN) for other agricultural, arid, and semiarid lands with AMC(II) and Ia=0.2S.

Cover Description		Curve Numbers for Hydrologic Soil Groups			
Cover Type	Hydrologic Condition	A	B	C	D
Other agricultural lands					
Pasture or range: continuous forage for grazing ^[a]	Poor	68	79	86	89
	Fair	49	69	79	84
	Good	39	61	74	80
Meadow (permanent) protected from grazing	Good	30	58	71	78
Brush: brush-weed-grass mixture with brush the major element ^[a]	Poor	48	67	77	83
	Fair	35	56	70	77
	Good	30	48	65	73
Woods-grass combination (orchard or tree farm) ^[b]	Poor	57	73	82	86
	Fair	43	65	76	82
	Good	32	58	72	79
Woods or forest land ^[c]	Poor	45	66	77	83
	Fair	36	60	73	79
	Good	30	55	70	77
Farmsteads		59	74	82	86
Arid and semiarid rangelands^[d]					
Herbaceous: mixture of grass, weeds, and low-growing brush (brush the minor element)	Poor	^[e]	80	87	93
	Fair	^[e]	71	81	89
	Good	^[e]	62	74	85
Oak-aspen-mountain brush	Poor	^[e]	66	74	79
	Fair	^[e]	48	57	63
	Good	^[e]	30	41	48
Pinyon, juniper or both; grass understory	Poor	^[e]	75	85	89
	Fair	^[e]	58	73	80
	Good	^[e]	41	61	71
Sagebrush with grass understory	Poor	^[e]	67	80	85
	Fair	^[e]	51	63	70
	Good	^[e]	35	47	55
Desert shrub	Poor	63	77	85	88
	Fair	55	72	81	86
	Good	49	68	79	84
^[a] Poor: < 50% ground cover Fair: 50 to 75% ground cover Good: > 75% ground cover ^[b] Computed for 50% woods and 50% grass; use CN for woods and pasture for other percentages. ^[c] Poor: forest litter, small trees, and brush are destroyed by heavy grazing or regular burning Fair: woods are grazed but not burned, and some forest litter covers the soil Good: woods are protected from grazing, and litter and brush adequately cover the soil ^[d] Poor: < 30% ground cover (litter, grass, and brush overstory) Fair: 30 to 70% ground cover Good > 70% ground cover ^[e] Curve numbers not developed. Source: SCS (1986).					